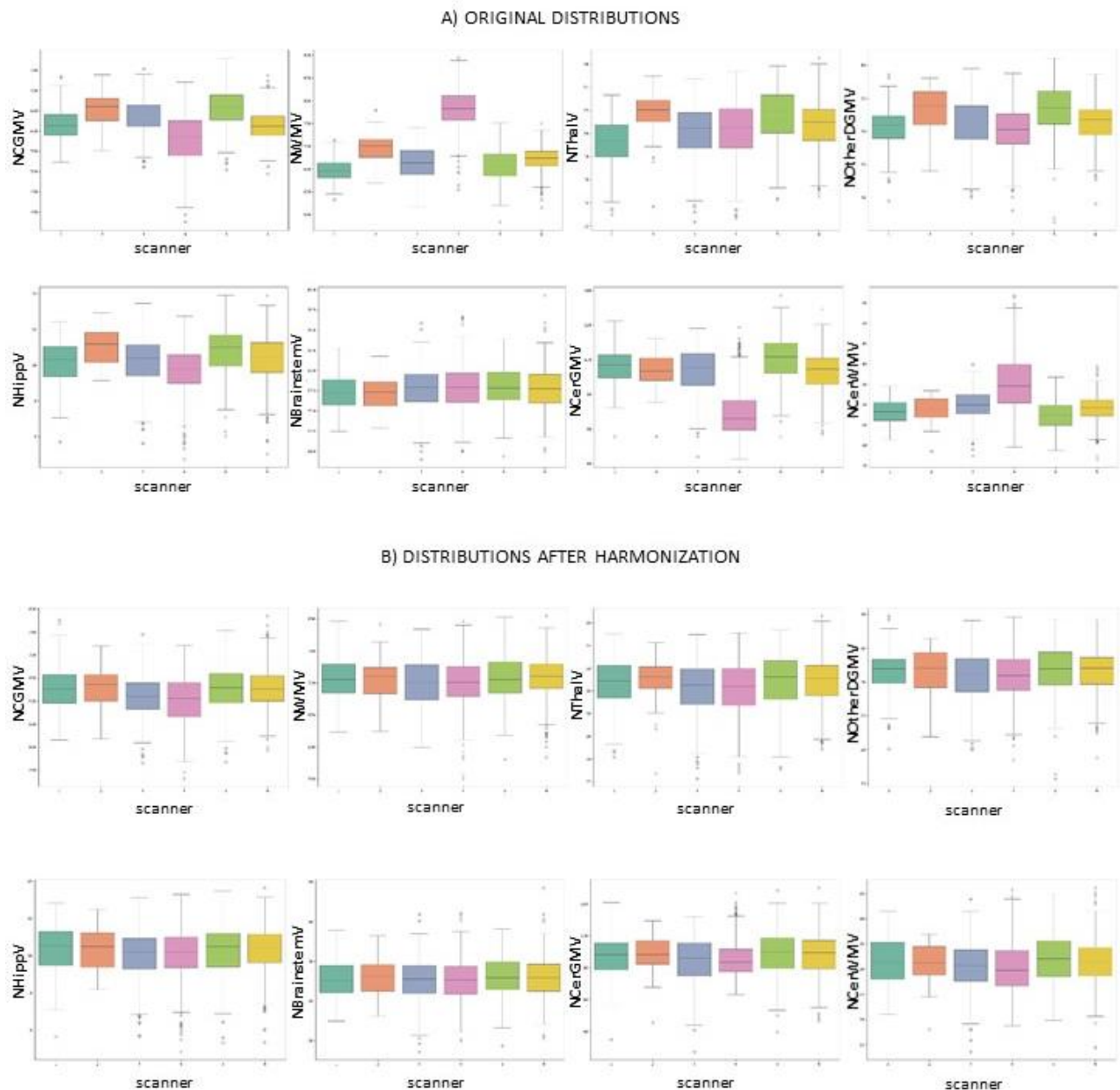


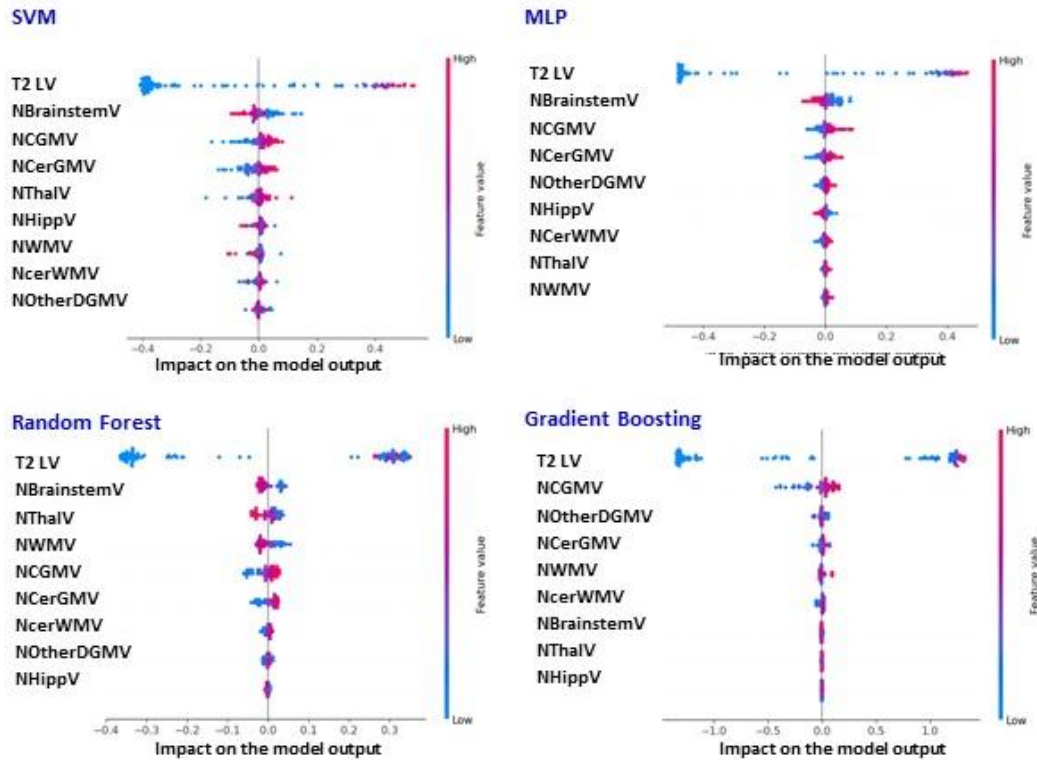
Supplementary Table 1. Parameters used for the acquisition of MRI sequences at the four participating centres.

	Milan		Naples		Rome	Siena
Scanner	Philips Achieva	Philips Ingenia	GE Signa HDxt	GE DV750	Siemens Magnetom Verio	Philips Achieva
Sequence for WM lesion assessment	DE TSE: TR=range 2599-2910 ms TE=16/80 ms FA=90° matrix=256x256 FOV=240x240 mm thickness=3 mm n. of slices=range 44-50 orientation=axial	3D FLAIR: TR=4800 ms TE=270 ms TI=1650 ms matrix=256×256 FOV=256×256 mm thickness=1 mm n. of slices=192 orientation=sagittal	DE TSE: TR=3080 ms TE=24/127.5 ms FA=90° matrix=256x384 FOV=240x240 mm thickness=3 mm n. of slices=44 orientation=axial	DE TSE: TR=5297 ms TE=18/93 FA=110° matrix=384x384 FOV=240x240 mm thickness=3 mm n. of slices=44 orientation=axial	DE TSE: TR=range 3320-5310 ms TE=10/103 ms FA=150° matrix=384x384 FOV=220x220 mm thickness=range 3-4 mm n. of slices=range 40-45 orientation=axial	DE TSE: TR=4000 ms TE=15/100 ms FA=90° matrix=240x240 (recon 352x352) FOV=240x240 mm thickness=3 mm n. of slices=44 orientation=axial
3D T1-weighted	TR=25 ms TE=4.6 ms FA=30° matrix=256x256 FOV=230x230x176 mm thickness=0.8 mm n. of slices=220 orientation=axial	TR=7.0 ms TE=3.2 ms TI=1000 ms FA=8° matrix=256x256 FOV=256x256x204 mm thickness=1 mm n. of slices=204 orientation=sagittal	TR=6.98 ms TE=2.85 ms TI=650 ms FA=8° matrix=256x256 FOV=256x256x199.2 mm thickness=1.2 mm n. of slices=166 orientation=sagittal	TR=6.9 ms TE=2.99 ms TI=650 ms FA=9° matrix=256x256 FOV=256x256x176 mm thickness=1 mm n. of slices=176 orientation=sagittal	TR=1900 ms TE=2.9 ms TI=900 ms FA=9° matrix=256x256 FOV=256x256x176 mm thickness=1 mm n. of slices=176 orientation=sagittal	TR=10 ms TE=3.9 ms TI=900 ms FA=8° matrix=256x256 FOV=256x256x192 mm thickness=1 mm n. of slices=192 orientation=axial

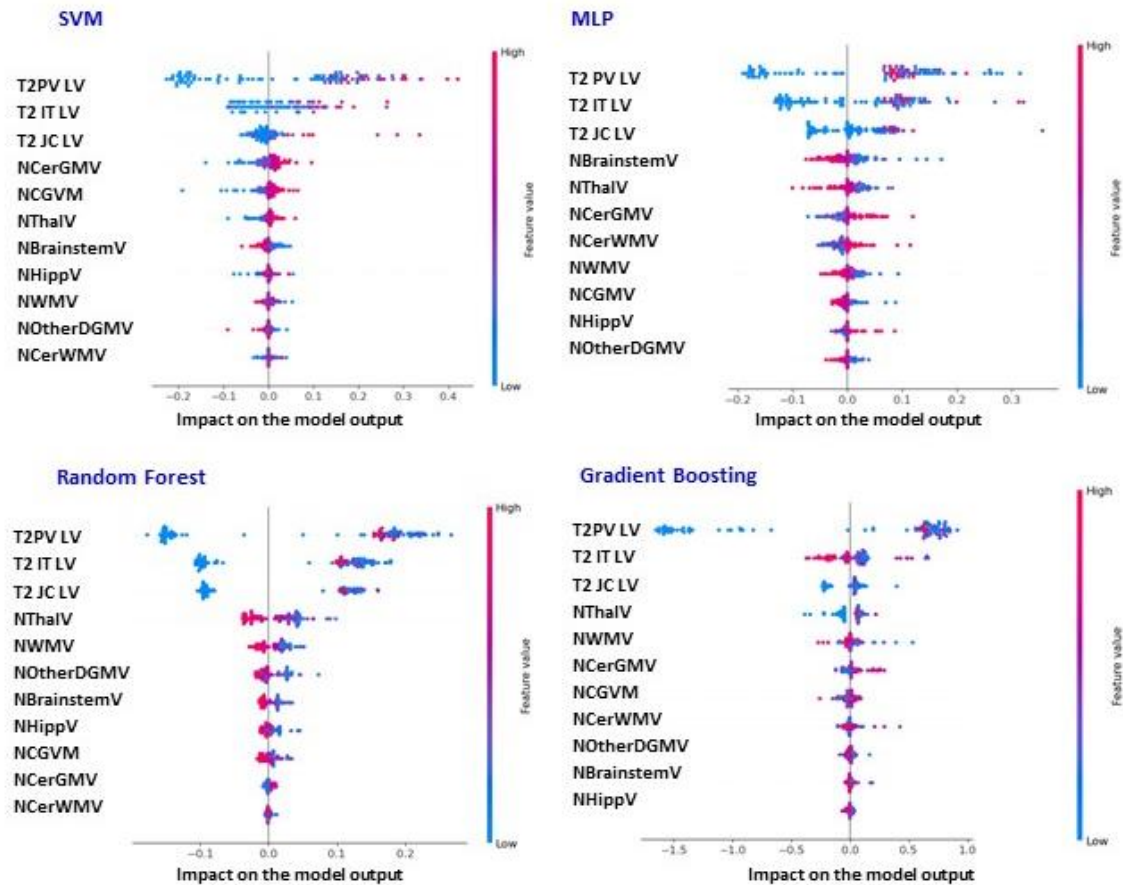
Abbreviations: DE TSE=dual echo turbo spin echo; FLAIR=fluid attenuated inversion recovery; WM=white matter; TR=repitition time; TE=echo time; TI=inversion time; FA=flip angle; FOV=field of view.



Supplementary Figure 1. Boxplots showing the distributions of MRI measures across scanners (scanner 1=Philips Achieva, University of Siena; scanner 2=GE Signa HDxt, University of Campania “Luigi Vanvitelli”; scanner 3=Philips Achieva, IRCCS San Raffaele Scientific Institute; scanner 4=Philips Ingenia, IRCCS San Raffaele Scientific Institute; scanner 5=GE DV750, University of Campania “Luigi Vanvitelli”; scanner 6=Siemens Magnetom Verio, Sapienza University of Rome) before (panel A) and after (panel B) harmonization performed using NeuroCombat procedure. Abbreviations: NCGMV=normalized cortical grey matter volume; NW/MV=normalized white matter volume; NThalV=normalized thalamic volume; NOtherDGMV=normalized volume of other deep grey matter nuclei; NHippV=normalized hippocampal volume; NBrainstemV=normalized brainstem volume; NCerGMV=normalized cerebellar grey matter volume; NCerWMV=normalized cerebellar white matter volume.



Supplementary Figure 2. SHAP-based feature contributions to SVM, MLP, Random Forest e Gradient Boosting classification performance (patients vs healthy controls) – Sensitivity Analysis I. Features best explaining the classification performances of support vector machine (SVM), multi-layer perceptron (MLP), Random Forest and Gradient Boosting models when distinguishing patients with multiple sclerosis (MS) from healthy controls (SHAP analysis) performed using only early MS patients (disease duration ≤ 5 years, sensitivity analysis I). Abbreviations: LV=lesion volume; NCGMV=normalized cortical grey matter volume; NWMV=normalized white matter volume; NThalV=normalized thalamic volume; NHippV=normalized hippocampal volume; NOtherDGMV=normalized volume of other deep grey matter nuclei; NCerGMV=normalized volume of cerebellar grey matter; NCerWMV=normalized volume of cerebellar white matter; NBrainstemV=normalized volume of brainstem.



Supplementary Figure 3. SHAP-based feature contributions to SVM, MLP, Random Forest e Gradient Boosting classification performance (patients vs healthy controls) – Sensitivity Analysis II. Features best explaining the classification performances of support vector machine (SVM), multi-layer perceptron (MLP), Random Forest and Gradient Boosting models when distinguishing patients with multiple sclerosis (MS) from healthy controls (SHAP analysis) performed using regional lesion volume metrics (sensitivity analysis II). Abbreviations: LV=lesion volume; PV=periventricular; JC=juxta-cortical; IT=infratentorial; NCGMV=normalized cortical grey matter volume; NWMV=normalized white matter volume; NThalV=normalized thalamic volume; NHippV=normalized hippocampal volume; NOtherDGMV=normalized volume of other deep grey matter nuclei; NCerGMV=normalized volume of cerebellar grey matter; NCerWMV=normalized volume of cerebellar white matter; NBrainstemV=normalized volume of brainstem.